

Member ID: _____

Time: _____

Rank: _____



SQL DATABASE FUNDAMENTALS (345)

STATE 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each) _____ (100 points)

TOTAL POINTS _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

MULTIPLE CHOICE

Identify the letter of the choice that *best* completes the statement or answers the question.

1. A database that follows 2NF has to first meet 1NF.
 - a. True
 - b. False
2. 1NF states that each table must have a primary key that is not a composite key.
 - a. True
 - b. False
3. One of the rules of 3NF is the database has no transitive functional dependencies. This means _____.
 - a. there are no composite keys
 - b. every child table has a foreign key that matches the size and datatype of the parent table
 - c. changing a non-key column, might cause any of the other non-key columns to change
 - d. the primary key is a numeric, auto-incremented key
4. When using GROUP BY on a Primary Key you would _____.
 - a. expect a much smaller dataset then if you didn't roll up the data
 - b. expect a dataset that is the exact same size as if you didn't roll up the data
 - c. not have any expectations without analyzing the data in the tables
 - d. not be able to interpret the data
5. When aggregating, the data will _____ come back sorted based on the grouped columns
 - a. always
 - b. never
 - c. sometimes
 - d. depend on further information to
6. To sort the Name field alphabetically from a-z and the Total field from largest to smallest the ORDER BY clause will be as follows: _____.
 - a. ORDER BY Name, Total
 - b. ORDER BY Name (a-z), Total (high-low)
 - c. ORDER BY Name ASC, Total ASC
 - d. ORDER BY Name, Total DESC

7. Because Indexes are used to speed up the execution of queries, DBAs will typically have multiple indexes on the key column of parent tables.
 - a. True
 - b. False

8. The command to list all tables in a given database is _____.
 - a. SHOW TABLES
 - b. LIST TABLES
 - c. MORE TABLES
 - d. ANALYZE TABLES

9. Given a table with field value, to find if that value that matches the contents of a list you would add the following to a WHERE clause: _____.
 - a. WHERE value LIKE ("A", "B", "C")
 - b. WHERE value IN ("A", "B", "C")
 - c. WHERE value IS GREATER THAN "A" and LESS THAN "C"
 - d. WHERE value BELONGS TO ("A", "B", "C")

10. Which of these is *not* a common use for an SQL trigger?
 - a. Enforce business rules
 - b. Cascading Deletes
 - c. Reducing complexity during development
 - d. Complex auditing to track table changes

Consider the following three tables for questions 11-19

Movie				MovieActor		Actor		
movieId	title	genreId	genre	movieId	actorId	actorId	name	dob
1	Speed	1	Horror	1	1	1	Bob	1950-01-01
2	Strength	2	War	1	3	2	Sally	1970-05-11
3	Size	4	Drama	1	6	3	Jane	2000-07-12
4	Shape	2	War	2	4	4	Carl	1970-05-11
				2	2	5	Jim	1999-06-01
				4	6	6	Sally	1994-06-24
				4	4			

11. How many records will the following query produce?

SELECT a.movieId FROM Movie a INNER JOIN MovieActor b ON a.movieId = b.movieId

- a. 8
- b. 7
- c. 6
- d. 5

12. How many records will the following query produce?

SELECT a.movieId FROM Movie a LEFT JOIN MovieActor b ON a.movieId = b.movieId

- a. 8
- b. 7
- c. 6
- d. 5

13. What would be the order of the actorId given the following query: `SELECT DISTINCT c.actorID FROM Movie a INNER JOIN MovieActor b ON a.movieId = b.movieId INNER JOIN Actor c ON b.actorId = c. actorId WHERE genreId = 2 ORDER by name ASC, dob DESC`
- a. 2, 4, 6
 - b. 6, 4, 2
 - c. 4, 6, 2
 - d. 2, 2, 6
14. Which query will produce a list of only the Movies that do not have any actors?
- a. `SELECT title FROM Movie a LEFT JOIN MovieActor b ON a.movieId = b.movieId WHERE b.movieId IS NULL`
 - b. `SELECT title FROM Movie a LEFT JOIN MovieActor b ON a.movieId = b.movieId WHERE a.movieId IS NULL`
 - c. `SELECT title FROM Actor a RIGHT JOIN MovieActor b ON a.movieId = b.movieId WHERE b.movieId IS NULL`
 - d. `SELECT title FROM Actor a RIGHT JOIN MovieActor b ON a.movieId = b.movieId WHERE a.movieId IS NULL`
15. It has been determined that *Sally* is also in the movie “*size*”. Which one of the following statements adds *Sally* as an actor for the movie.
- a. `UPDATE MovieActor SET movieId = 3, actorId = 6`
 - b. `INSERT INTO MovieActor values (3, 6)`
 - c. `ADD TO MovieActor (movieId, actorId) VALUES (3, 6)`
 - d. `CREATE ROW IN MovieActor (movieId, actorId) VALUES (3, 6)`
16. What is the first normalization form that the Movie table fails?
- a. 1NF
 - b. 2NF
 - c. 3NF
 - d. 4NF
17. The Actor table fails which database normalization form?
- a. 1NF
 - b. 2NF
 - c. 3NF
 - d. The Actor table does not fail any normalization forms

18. Which query will produce the movie(s) with the oldest actors / actresses?
- a. SELECT title, dob FROM Movie a INNER JOIN Actor b ON a.movieId = b.movieId WHERE dob = MAX(dob)
 - b. SELECT title, dob FROM Movie a INNER JOIN MovieActor b ON a.movieId = b.movieId INNER JOIN Actor c ON b.actorId = c.actorId WHERE dob = MAX(dob)
 - c. SELECT title, dob FROM Movie a INNER JOIN MovieActor b ON a.movieId = b.movieId INNER JOIN Actor c ON b.actorId = c.actorId WHERE dob in (SELECT MAX(dob) FROM Actor)
 - d. SELECT title, dob FROM Movie a INNER JOIN MovieActor b ON a.movieId = b.movieId INNER JOIN Actor c ON b.actorId = c.actorId WHERE dob = (NOW()-dob)
19. The Movie table is known as a _____ table because it only contains foreign keys that allow two other tables to join to each other.
- a. mapping
 - b. composite
 - c. interval
 - d. Boyce Codd
20. Given two tables, table1 and table2, CROSS JOIN produces _____.
- a. Every row where the key used in the join is in both tables
 - b. Every row where the key used in the join is in table1, regardless of whether or not it is in table2
 - c. Every row where the key used in the join is in table2, regardless of whether or not it is table 1
 - d. Every row from table1 joined with each row in table2
21. Which would *not* be a reason to create a database view to present to your users?
- a. Views always make accessing data more efficient
 - b. To hide Complexity
 - c. To protect specific data from specific users
 - d. They can replace tables to support legacy code

Consider the table *category* with the following fields *id*, *item*, *year*, *description* in the database *lift* for questions 22-25

22. The *year* field is an integer and we are only storing the years it takes to disintegrate.

Given a max number of years of 2999, what is the smallest data type to hold that value?

- a. Blob
- b. Tinyint
- c. SmallInt
- d. MediumInt

23. The *description* field is currently a varchar(150). How much space is taken up for this field when the data in the field is only 50 bytes?

- a. 150
- b. 75
- c. 52
- d. 50

24. How would you add an index to the *category* table on the *item* field?

- a. ADD INDEX index1 ON category item
- b. CREATE INDEX index1 ON category (item)
- c. ALTER TABLE category ADD INDEX item
- d. USE category ADD INDEX index1 ON item

25. How would you remove the *category* table?

- a. DELETE FROM lift WHERE table_name = "category"
- b. REMOVE TABLE category
- c. DROP TABLE category
- d. ERASE TABLE category

26. A _____ is a named and optionally parameterized compiled set of SQL database access statements that can be executed as a unit.

- a. Parameterized Function
- b. Dynamic Linked Library
- c. Application Programming Interface
- d. Stored Procedure

27. To combine the data from two separate queries one could use a _____.
a. UNION
b. SUBQUERY
c. FULL OUTER JOIN
d. INTERSECTION
28. Standard properties of a transaction are often referred to by the acronym _____.
a. CARD
b. ACID
c. DARC
d. DAC
29. What is the purpose of the “C” property, Consistency, in transaction processing?
a. Only valid data following all rules and constraints is written in the database
b. Enabled transactions to operated independently of each other
c. Ensures the result of a committed transaction is permanent in case of a system failure
d. Ensures that all operations are completed successfully
30. What is the purpose of the “A” property, Automicity, in transaction processing?
a. Only valid data following all rules and constraints is written in the database
b. Enabled transactions to operated independently of each other
c. Ensures the result of a committed transaction is permanent in case of a system failure
d. Ensures that all operations are completed successfully
31. Which of the following is not a commonly used column constraint in a database?
a. UNIQUE
b. NOT NULL
c. FOREIGN
d. MAX
32. A primary key can allow duplicates in special circumstances?
a. True
b. False

33. When designing a child table, the foreign key column(s) in the child table must have the same name as the primary key column(s) in the parent table.
- True
 - False
34. When changing a column's data type, which of the following is a true statement?
- varchar(50) to char(20) may result in data loss
 - Int to tinyint will cause an error
 - Int to char will cause an error
 - Int to smallint will result in data loss
35. A database will experience a deadlock when _____.
- Two transactions try to update the same record
 - Transaction one locks table1 but needs table2; transaction two locks table2 but needs table1
 - A database is being maintained while a transaction is trying to update a table
 - Transaction one is waiting on transaction two to be rolled back after a system failure.

Consider the following for questions 37-44 (Added in MySQL)

```
DROP TABLE IF EXISTS testTable;
```

```
CREATE TABLE `testTable` (  
    `id` INT NOT NULL AUTO_INCREMENT,  
    `fName` VARCHAR(45) NOT NULL,  
    `lName` VARCHAR(45),  
    `dob` DATETIME NOT NULL,  
    PRIMARY KEY (`id`),  
    UNIQUE INDEX `id_UNIQUE` (`id` ASC) VISIBLE,  
    INDEX `name` (`lName` ASC) VISIBLE);
```

```
INSERT INTO testTable (fname, dob) VALUES ("Bob", "1980-01-01");
```

```
DELETE FROM testTable WHERE id > 0;
```

```
INSERT INTO testTable (fname, lname, dob) VALUES ("Joe", "Smith", "1990-10-10");
```

```
INSERT INTO testTable VALUES (12, "Jane", "Jones", "1985-11-11");
```

```
INSERT INTO testTable (fname, lname, dob) VALUES ("Joe", "Jones", "1990-10-10");
```

```
INSERT INTO testTable VALUES (24, "Alice", "Anders", "2000-05-05");
```

```
INSERT INTO testTable (fname, dob) VALUES ("Bob", "1980-01-01");
```

36. SELECT id FROM testTable will return _____.

- a. Null, 12, null, 24, null
- b. 1, 12, 2, 24, 3
- c. 2, 12, 13, 24, 25
- d. 1, 12, 13, 24, 25

37. What will SELECT COUNT(DISTINCT fName), MAX(dob) FROM testTable return?

- a. 4, 2000-05-05
- b. 5, 2000-05-05
- c. 5, 1980-01-01
- d. 4, 1985-01-01

38. How many rows will the following query produce - SELECT 5*6+4 FROM testTable WHERE true?

- a. 0
- b. 1
- c. 5
- d. 6

39. How many rows will the following query produce: SELECT fName, count(id) FROM testTable GROUP BY fName?

- a. 1
- b. 3
- c. 4
- d. 5

40. The following SQL statement - INSERT INTO testTable VALUES (24, "Ginny", "Gerber", "2000-04-04") will have _____ as a result.

- a. The table now has 6 rows, the auto-increment will take care of creating a new id.
- b. The table will continue to have 5 rows and the row with id 24 will be overwritten
- c. The table will have 6 rows, the id will be 24, different names assure uniqueness
- d. An Error be will thrown

41. The following SQL statement - INSERT INTO testTable (lName, dob) VALUES ("Lincoln", "2006-01-01") will have _____ as a result:

- a. A new row will be added to the end of the table with a new id which is one larger than the last id created, null first name, Lincoln last name, and dob of 2006-01-01
- b. A new row will be added to the end of the table with a null id, null first name, Lincoln last name, and dob of 2006-01-01
- c. A new row will be inserted into the middle of the table with a new id which fills in with an unused id, Lincoln last name, and dob of 2006-01-01
- d. An Error will be thrown

42. What will be the result set of the following query - `SELECT fName, count(id) as idCount FROM testTable GROUP by fName HAVING idCount > 1?`
- a. Joe, 1
Joe, 1
 - b. Joe, 2
 - c. Joe, 2
Bob, 2
 - d. An error will be thrown
43. What is the difference between the following two queries?
- `SELECT fName FROM testTable WHERE id > 15`
- `SELECT fName FROM testTable HAVING id > 15`
- a. The two queries produce identical result
 - b. Having is only applies with a GROUP BY - second query produces no result
 - c. The query with the WHERE clause will error because id is not in the field list
 - d. The query with the HAVING clause will error because id in not in the field list
44. When writing applications that use a database, it is customary to use the “root” user so the application can accomplish the tasks it needs to accomplish.
- a. True
 - b. False
45. In database terminology what is a sandbox?
- a. The process of ensuring the database passes all Normalization forms
 - b. A development database separate from production
 - c. The name given to a multi-user database
 - d. The name given to a database backup
46. What holds true for the following SQL to be valid - `SELECT theName FROM firstDB.table1 UNION SELECT theTitle FROM secondDB.table2?`
- a. theName must be a compatible data type with theTitle
 - b. theName must be the same data type as theTitle
 - c. Query can not run because column names must be the same
 - d. Query can not run because tables are from different databases

47. Which of the following is *not* a standard key type on a database table?
- a. Candidate Key
 - b. Primary Key
 - c. Associate Key
 - d. Alternate Key
48. Production databases automatically maintain referential integrity between tables using primary and foreign keys.
- a. True
 - b. False
49. Like a standard if statement, a WHERE clause evaluates in the order in which it is presented.
- a. True
 - b. False
50. A duty a DBA is *not* typically in charge of would be _____.
- a. designing stored procedures
 - b. creating databases
 - c. capacity planning
 - d. scheduling database backups